

AP Calculus II
Notes 8.1
Basic Integration Techniques

In this section, we will review all the form-fitting integration techniques we have learned to get a firm grasp on when to use; which to not only get them correct, but to do so quickly and efficiently.

Ex. 1: Evaluate the following closely related integrals:

a) $\int \frac{4}{x^2 + 9} dx$

b) $\int \frac{4x}{x^2 + 9} dx$

c) $\int \frac{4x^2}{x^2 + 9} dx$

It is VERY important to know all the integration rules inside and out so you know which integrals apply to the certain rules. Do NOT force a problem into a rule; it must fit perfectly.

Ex. 2: Evaluate $\int_0^1 \frac{x+3}{\sqrt{4-x^2}} dx$.

Ex. 3: Find the value of k such that the area under the graph of $f(x) = \frac{e^x}{1+e^x}$ from $x = 0$ to $x = k$ is $\ln 3$.

Ex. 4: The rate in which snow is falling can be modeled by the equation $R(t) = \frac{\cos^2 t + \sin t}{\cos t}$ for $t \geq 0$. How much snow has fallen on the ground from $0 \leq t \leq \frac{\pi}{4}$?

Ex. 5: A particle moves along the x -axis such that its velocity can be modeled by the function $v = t \left(1 + \frac{1}{t}\right)^2$ for $t > 0$. Find the position of the particle at $t = 3$ given that $x(1) = 4$.

Summary of Procedures to Fit Integrands Into Integration Rules

Ex. $(1 + e^x)^2$ **1)**

Ex. $\frac{1+x}{x^2+1}$ **2)**

Ex. $\frac{3}{5+2x+x^2}$ **3)**

Ex. $\frac{x^2}{x^2+1}$ **4)**

Ex. $\cot^2 x$ **5)**

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Notes 8.2
Integration by Parts

In this section, we now look at a new technique that is very useful for integrands involving products of algebraic and transcendental functions. Integration by parts is based on the formula for the derivative of a product:

$$\frac{d}{dx}[uv]=$$

If we integrate both sides of this equation, we get:

By rewriting this equation, you obtain the following integration rule:

Theorem – Integration by Parts

If u and v are functions of x and have continuous derivatives, then

Guidelines for Integration by Parts:

- 1) Try letting dv be the most complicated portion of the integrand that fits a basic integration rule.
- 2) Try letting u be the portion of the integrand whose derivative is a function **simpler** than u .
- 3) The dv always includes the dx of the original integrand.

A little help for picking which part is u

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Ex. 1: Integrate the following using Integration by Parts when necessary:

a) $\int 3x^2 \ln x \, dx$

b) $\int x e^{\frac{2x}{3}} \, dx$

Ex. 3: The acceleration of a particle moving along the x -axis is modeled by $a(t) = t^2 \cos t$ for all $t \geq 0$. Write the velocity equation for any time t given $v(0) = -2$.

Ex. 3: Find the area of the region bounded by the x -axis, the graph $y = \arcsin x$ and $x = 0$ and $x = \frac{1}{2}$.

Ex. 4: Let f be a differentiable function such that $\int f(x) \sin x \, dx = -f(x) \cos x + \int 4x^3 \cos x \, dx$. Which of the following could be $f(x)$?

(A) $\sin x$

(B) $4x^3$

(C) $-x^4$

(D) x^4

Ex. 5:

x	1	2	3	4	5
$f'(x)$	62	30	20	15	12

The function f is twice differentiable for $x \geq 1$ where $f(5) = -6$. Selected values of the positive and decreasing function f' , the derivative of f , are given in the table above.

If $\int_1^5 x f''(x) \, dx = -100$, find $f(1)$.

And now, the moment you've all been waiting for, Mr. Gough's favorite problem in all of calculus!

Ex. 6: Evaluate $\int e^x \sin x \, dx$

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Notes 8.5
Partial Fractions

This section takes a look at another integration technique in which a rational function is decomposed into simpler rational functions to make the integration process easier. Before we actually integrate, let's check out how to decompose rational functions.

Ex. 1: Write the partial fraction decomposition for $\frac{4x+7}{x^2+x-6}$.

Ex. 2: Write the partial fraction decomposition for $\frac{5x^2+20x+}{x^3+3x^2+2x}$.

Ex. 3: Evaluate $\int \frac{3x - 1}{x^2 - x - 6} dx$.

Ex. 4: Evaluate $\int \frac{21}{2x^2 + x - 6} dx$

Ex. 5: The population of Haverford is growing at a rate modeled by $G(t) = \frac{t^2+3t-30}{t^2-25}$ where $G(t)$ is millions of people per year and t is years since 2010 ($t = 0$). If there are 123 million people in the year 2011, what will the population be in 2014?

Ex. 6: $\int \frac{12}{(x-1)(x-5)} dx =$

- (A) $-3\ln|x-1| + 3\ln|x-5| + C$
- (B) $-2\ln|x-1| + 2\ln|x-5| + C$
- (C) $3\ln|x-1| - 3\ln|x-5| + C$
- (D) $12\ln|x-1| + 12\ln|x-5| + C$

Ex. 7: Evaluate $\int \frac{2x^2 - 6x - 18}{x^3 - 9x} dx$

Ex. 8: Describe what process and which formula you would use, if any, to integrate.

a) $\int \frac{4}{x^2 + 10x + 30} dx$

b) $\int \frac{4x + 6}{x^2 + 3x - 1} dx$

c) $\int \frac{x^2 + 9x - 15}{2x} dx$

d) $\int \frac{4}{x^2 + 10x + 24} dx$

e) $\int \frac{4x^3 + 31}{x^2 + 10} dx$

f) $\int \frac{4 \cos x}{\sin^2 x + 6 \sin x + 8} dx$

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Notes 8.7
Indeterminate Forms/L'Hôpital's Rule

Review of Limits:

Evaluate the following limits:

a) $\lim_{x \rightarrow 4} \frac{x^2 - x - 12}{x^2 - 16}$

b) $\lim_{x \rightarrow \infty} \frac{3x^2 - x - 12}{4x^2 - 16}$

c) $\lim_{t \rightarrow \frac{\pi}{2}} \frac{1 - \sin^2 t}{t \cos t}$

d) $\lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h}$

Now, what about $\lim_{x \rightarrow 0} \frac{e^x - 1}{\sin x}$? If you can't figure out a way to simplify this, then use the table/graph to evaluate the limit.

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{\sin x} =$$

L'Hôpital's Rule:

Let f and g be functions that are differentiable on an open interval (a, b) with c , except possibly at c .

If the $\lim_{x \rightarrow c} f(x) = 0$ and $\lim_{x \rightarrow c} g(x) = 0$, then $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$ is indeterminate and $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$

This result also applies if the $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$ is some other indeterminate form $\frac{\pm\infty}{\pm\infty}$, $\pm\infty \mp \infty$, $0 \cdot \infty$, etc...

There are three major mistakes that are made in L'Hôpital's Rule:

a)

b)

c)

Ex. 1: Evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1}{\sin x}$

Ex. 2: Evaluate the following limits:

a) $\lim_{x \rightarrow 4} \frac{x^2 - x - 12}{x^2 - 16}$

b) $\lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos x}{h}$

c) $\lim_{x \rightarrow \pi} \frac{\cos x + \sin(2x) + 1}{x^2 - \pi^2}$ is

(A) $\frac{1}{2\pi}$

(B) $\frac{1}{\pi}$

(C) 1

(D) nonexistent

Limits to Infinity

Ex. 3: Evaluate the following limits:

a) $\lim_{x \rightarrow -\infty} \frac{3x^2 + 7x - 5}{5 - 4x^2}$

b) $\lim_{t \rightarrow \infty} t^2 e^{-2t}$

Ex. 4: Evaluate the following limits:

a) $\lim_{p \rightarrow 5} \frac{|p-5|}{p-5}$

b) $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 + 5}}{x + 1}$

Ex. 5: Evaluate the following limits without a calculator:

a) $\lim_{x \rightarrow \pi} \frac{\sin x}{x}$

b) $\lim_{x \rightarrow \pi} \frac{\sin x}{x - \pi}$

c) $\lim_{x \rightarrow \pi} \frac{\int_{\pi}^x \sin t \, dt}{x - \pi}$

d) $\lim_{x \rightarrow 0} \frac{\cos^2 x - 1}{e^{3x} - 3x - 1}$

e) $\lim_{x \rightarrow 0} \frac{7x - \sin x}{x^2 + \sin(3x)} =$

f) $\lim_{x \rightarrow \infty} \frac{5x^{100}}{e^x}$

(A) 6

(B) 2

(C) 1

(D) 0

g) If $f(x) = (x^3 + 1)^2$, find $\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1}$

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Notes 8.8
Improper Integrals

The definition of a definite integral $\int_a^b f(x)dx$ requires that the interval $[a, b]$ be finite. Also, the FTC requires that the function is continuous on $[a, b]$ in order to evaluate the antiderivative. Here, we will look at a procedure for evaluating integrals that does not satisfy any of these requirements. Integrals that have either one or both of the limits of integration which are infinite or have a finite number of discontinuities are known as **improper integrals**.

Improper Integrals With Infinite Integration Limits:

1. If f is continuous on the interval $[a, \infty)$, then
$$\int_a^{\infty} f(x)dx = \lim_{b \rightarrow \infty} \int_a^b f(x)dx$$
2. If f is continuous on the interval $(-\infty, b]$, then
$$\int_{-\infty}^b f(x)dx = \lim_{a \rightarrow -\infty} \int_a^b f(x)dx$$
3. If f is continuous on the interval $(-\infty, \infty)$, then
$$\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^c f(x)dx + \int_c^{\infty} f(x)dx$$

In the first two cases, the improper integral **converges** if the limit exists, otherwise the improper integral **diverges**. In the third case, if either of the two diverge, then the original integral diverges.

Ex. 1: Evaluate the following integral:
$$\int_1^{\infty} \frac{dx}{\sqrt[5]{x}}$$

Ex. 2: Evaluate the following improper integrals:

a) $\int_1^{\infty} \frac{dx}{2x+1}$

b) $\int_{-\infty}^{\infty} \frac{dt}{4+t^2}$

Ex. 3: Evaluate $\int_1^{\infty} \left(\frac{1}{p+4} - \frac{1}{p+6} \right) dp$

Improper Integrals With Infinite Discontinuities:

1. If f is continuous on the interval $[a, b)$ and has a discontinuity at b , then $\int_a^b f(x)dx = \lim_{c \rightarrow b^-} \int_a^c f(x)dx$
2. If f is continuous on the interval $(a, b]$ and has a discontinuity at a , then $\int_a^b f(x)dx = \lim_{c \rightarrow a^+} \int_c^b f(x)dx$
3. If f is continuous on the interval $[a, b]$ except at c , then $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$

In the first two cases, the improper integral **converges** if the limit exists, otherwise the improper integral **diverges**. In the third case, if either of the two diverge, then the original integral diverges.

Ex. 4: Evaluate the following improper integrals:

a) $\int_0^2 \frac{dn}{(n-2)^3}$

b) $\int_{-1}^8 \frac{dx}{\sqrt[3]{x}}$

Ex. 5: A solid is formed by revolving, around the x -axis, the unbounded region lying between the x -axis and the graph $f(x) = \begin{cases} 6 - x & 1 \leq x \leq 4 \\ \sqrt{xe^{4-x}} & x > 4 \end{cases}$. Find the volume of the solid formed given $x \geq 1$.

Ex. 6:

x	1	2	3	4	5
$f'(x)$	62	30	20	15	12

The function f is twice differentiable for $x \geq 1$. Selected values of the positive and decreasing function f' are given in the table above. The graph of f' has a horizontal asymptote at $y = 0$.

a) Evaluate $\int_1^{\infty} f''(x) \sin(f'(x)) dx$.

b) Evaluate $\int_1^5 \left(\frac{f''(x)}{\sqrt{f'(x) - 12}} \right) dx$.